

KHALID SHAMS

Senior Software Engineer · .NET & Azure · Solution Architect

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PROFESSIONAL SUMMARY

Senior .NET engineer and solution architect with 16+ years building and operating API-driven systems. Hands-on across C#/ASP.NET Core, Azure API Management, Bicep infrastructure-as-code, Docker, and CI/CD. Currently designs, builds, deploys, and operates three commercial systems in production single-handedly — a multi-tenant SaaS, an event-driven LLM pipeline, and a storefront platform — covering architecture, containerized deployment, observability, and on-call. Previously led a 30+ engineer and QA team delivering utility systems serving 100k+ users and 1M+ monthly transactions, cutting manual reporting effort by 65%. Uses Claude Code and comparable AI-assisted tooling in daily engineering work.

SELECTED PROJECTS

Coupon Service — .NET 8 behind Azure API Management, deployed from an Azure DevOps pipeline *self-directed build*

- Built and deployed a coupon and ordering service end to end: .NET 8 API behind Azure API Management, React frontend with MSAL, all infrastructure in Bicep, and a six-stage Azure DevOps YAML pipeline that provisions and deploys the whole system.
- Verified the from-scratch claim rather than asserting it — deleted the resource group and rebuilt from empty in 17 minutes, which surfaced a purge step that had been broken across eleven green runs because its branch had never executed.
- Made the system fully passwordless: managed identities between every service, and Azure SQL reached without a tenant-admin Directory Readers grant by supplying the identity's client ID as a contained-user SID.
- Enforced OAuth2 authorization-code-with-PKCE at the gateway on order submission only, with audience, issuer and scope pinned to claims observed in a real token; the backend accepts requests from the gateway identity alone and is unreachable directly.
- Modelled coupon rules behind a clean interface with no access to prices, made client-supplied prices structurally impossible to honour, and covered both with Reqnroll BDD scenarios verified by deliberately breaking the server to confirm the tests fail.

Azure API Management · Azure Bicep · Azure DevOps Pipelines (YAML) · Microsoft Entra ID · Managed Identity · OAuth2 / PKCE · Azure SQL · App Service · Application Insights · Reqnroll · React · MSAL

AI Job-Search Copilot — Event-driven LLM pipeline with a polyglot microservice *jobcopilot.dentflowbd.com · source on GitHub*

- Built and operate a seven-container distributed system across two bounded contexts, with GitHub Actions building four images and deploying on push behind a deploy key pinned to a forced command.
- Run a production LLM workload: an ASP.NET Core API publishes to RabbitMQ rather than calling the model inline, a C# BackgroundService scores against Google Gemini, and results are pushed to the browser over SignalR using per-user groups keyed on the JWT subject claim.
- Replaced a shared queue with a fanout exchange after determining a second consumer would have silently round-robin away half of all live SignalR pushes; verified against per-queue message statistics before and after.
- Mitigated prompt injection in two independent layers — delimited, data-not-instructions framing on input and clamped, length-capped output — and tested with a real injection attempt.
- Adopted MongoDB Atlas under Terraform by import rather than re-creation, using plan to surface divergence from live reality; Grafana Alloy ships metrics and logs to Grafana Cloud.

ASP.NET Core 8 · C# · RabbitMQ · SignalR · React · TypeScript · Node.js · PostgreSQL · MongoDB Atlas · Terraform · Grafana Cloud · Docker Compose · GitHub Actions · Google Gemini

DentalPMS — Multi-tenant SaaS practice management for dental clinics *demo.dentflowbd.com/login*

- Designed, built, deployed, and operate a commercial platform with paying clinics on it: 127k lines of C#, 41 domain entities, 167 automated tests, 33 EF Core migrations.
- Enforced tenant isolation structurally through EF Core global query filters at the DbContext, so a cross-tenant query cannot be written by accident rather than relying on developer discipline.
- Clean Architecture across four projects; Blazor Server UI over 81 components on PostgreSQL; JWT in HttpOnly cookies with policy-based authorization; nginx on a Linux VPS with per-clinic custom domains and 13 written runbooks.

ASP.NET Core 8 · Blazor Server · C# · PostgreSQL · EF Core · ASP.NET Identity · Docker · nginx · xUnit · Playwright

One-Page Commerce — Single-product Cash-on-Delivery storefront platform *one-page-commerce.vercel.app*

- Built and operate a commercial storefront platform with live stores selling through it, including an admin panel a non-developer can run unsupervised.
- Recompute every checkout price, discount, and delivery fee server-side from the database, so a client-posted total is never trusted; single-tenant and pooled multi-tenant deployments run from one codebase.

Next.js (App Router) · TypeScript · PostgreSQL · Tailwind CSS · Docker Compose

PROFESSIONAL EXPERIENCE

Solution Delivery Specialist — Today's Tech

Apr 2025 – Present

- Own end-to-end solution architecture — from discovery and estimation through architecture alignment, implementation, quality gates, and production release.
- Translate business goals into technical roadmaps, system designs, release plans, and measurable delivery outcomes.

- Set technical direction across product, engineering, QA, and operations to keep enterprise delivery predictable and transparent.
- Own sprint planning, backlog prioritization, dependency management, delivery risk, and release readiness.

Technical Project Manager — BJIT Ltd

Oct 2016 – Mar 2025

- Architected the CWASA Digital Ecosystem as project lead — billing, non-revenue-water analysis, management dashboards, predictive analytics, SCADA and GIS integration, and reporting workflows.
- Delivered API-gateway-fronted microservices over ASP.NET Core and SQL Server supporting 100k+ users, 1M+ monthly transactions, and a 65% reduction in manual reporting effort.
- Led a 30+ member cross-functional team of developers and QA engineers across planning, implementation, review, and release; mentored engineers through architecture and code review.
- Conducted architecture reviews, implementation guidance, dependency management, and production-readiness planning.

Software Engineer — Intelligent Image Management Inc.

Jun 2015 – Oct 2016

- Implemented OCR and intelligent data-capture solutions using the Tesseract OCR engine, with image-processing and extraction workflows for scanned and image-based inputs.

Software Engineer — Simplexhub Ltd

Aug 2013 – Jun 2015

- Developed e-commerce platforms with integrated payment gateways and secure transaction workflows, plus medical service management solutions and backend services for business-facing applications.

Software Engineer — LEADS Corporation Ltd

Jul 2012 – Aug 2013

- Developed financial analytics tools for stock-market data analysis, with real-time processing workflows and predictive logic for market-trend decision support.

Lab Technical Officer — BRAC University

Jan 2012 – Jul 2012

- Supported academic programming labs, maintained network and server infrastructure, and provided technical guidance to students and faculty.

Programmer (.NET) — CACTS Ltd

Apr 2010 – Dec 2011

- Developed core ERP modules from scratch across inventory, HR, payroll, and administrative workflows, with relational-database-backed business processes and reporting modules.

TECHNICAL SKILLS

Azure & Cloud: Azure API Management · Azure Bicep (IaC) · Azure DevOps Pipelines (YAML) · Microsoft Entra ID · Azure App Service · Azure SQL · Application Insights · Azure CLI / ARM · Managed Identity · Workload Identity Federation · AWS Lambda · MongoDB Atlas

Languages & Frameworks: C# · ASP.NET Core · ASP.NET · Blazor Server · MVC · ADO.NET · WCF · Node.js · TypeScript · JavaScript/ES6 · HTML5 · CSS3

APIs & Integration: REST · Web API · OpenAPI / Swagger · OAuth2 · PKCE · JWT · SOAP · API Gateway patterns (Azure APIM, Ocelot) · Webhooks · Rate limiting & quotas

AI & LLM: Production LLM pipelines · Google Gemini API · Prompt-injection mitigation · AI-assisted development workflows (Claude Code and comparable tooling)

Messaging & Real-Time: RabbitMQ · SignalR · Event-Driven Architecture · Background Workers · WebSockets

Architecture & Patterns: Clean Architecture · Multi-Tenant SaaS · Microservices · Domain-Driven boundaries · Repository · Factory · Singleton

Databases & ORM: PostgreSQL · MS SQL Server · Oracle · MySQL · MongoDB · Entity Framework Core · Entity Framework · Dapper · NHibernate

DevOps & Infrastructure: Docker · Docker Compose · Azure Bicep · Terraform · Azure DevOps Pipelines · GitHub Actions · Jenkins · nginx · Linux VPS Operations · TLS / Let's Encrypt · Git · TFS · SVN

Testing & Quality: xUnit · NUnit · MSTest · Reqnroll (BDD) · Playwright · Vitest · React Testing Library · Postman · Fiddler

Observability: Grafana Cloud · Grafana Alloy · Structured Logging · Liveness / Readiness Probes · Container Healthchecks

Frontend: React.js · Next.js · Angular · Blazor · Tailwind CSS · TanStack Query · Material-UI · Bootstrap · jQuery

EDUCATION & CERTIFICATION

Bachelor of Science in Computer Science — BRAC University

Certified ScrumMaster® — Scrum Alliance® | Dec 2023 – Dec 2025 (not currently renewed)

Languages: Bengali (Native) | English (Fluent)